

ADAK, AK, USA

WMO: 704540

Lat: 51.883N

Lon: 176.650W

Elev: 5

StdP: 101.26

Time Zone: -10.00 (NAH)

Period: 94-19

WBAN: 25704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -4.9	(d) -12.0	(e) 1.3	(f) -5.1	(g) -10.1	(h) 1.6	(i) -3.4	(j) 20.1	(k) 1.4	(l) 18.3	(m) 2.1	(n) 1.8	(o) 230	(p) 0.895

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.4	(c) 15.9	(d) 13.6	(e) 14.0	(f) 11.8	(g) 12.9	(h) 10.8	(i) 14.0	(j) 15.4	(k) 12.5	(l) 13.6	(m) 11.5	(n) 12.7	(o) 6.8	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 13.0	(b) 9.3	(c) 14.5	(d) 12.1	(e) 8.8	(f) 13.8	(g) 10.9	(h) 8.1	(i) 12.5	(j) 39.0	(k) 15.2	(l) 35.7	(m) 13.6	(n) 32.5	(o) 12.6	(p) 19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.7	14.1	-10.8	18.3	2.0	1.5	-12.3	19.3	-13.5	20.2	-14.6	21.0	-16.1	22.0
			-11.5	15.9	2.0	1.9	-12.9	17.2	-14.1	18.3	-15.3	19.3	-16.7	20.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	0.3	1.1	1.3	2.7	4.9	7.6	9.8	11.1	9.4	6.3	3.4	1.1		
	DBStd	4.22	3.01	2.41	2.35	1.64	1.48	1.37	1.50	1.73	1.83	2.08	2.26	2.60		
	HDD10.0	1919	300	249	271	218	159	72	22	6	31	117	198	276		
	HDD18.3	4891	558	483	529	468	417	321	265	225	268	374	448	534		
	CDD10.0	70	0	0	0	0	0	1	15	39	13	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.4	6.5	7.7	7.1	6.7	6.0	5.2	4.9	5.2	6.5	6.8	7.1	7.2		
(15) Precipitation	PrecAvg	1435	136	101	126	107	78	64	71	102	130	152	184	170		
	PrecMax	1774	261	208	245	258	152	124	155	186	287	254	296	258		
	PrecMin	949	69	50	59	36	16	23	18	32	63	66	75	98		
	PrecStd	209	44	37	46	47	33	24	32	37	49	47	58	45		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	7.4	7.4	7.8	10.5	13.6	17.2	18.3	15.9	12.6	11.0	8.2		
		MCWB	5.4	5.8	5.7	5.7	7.4	10.6	13.9	16.7	13.1	9.9	9.4	6.5		
	2%	DB	6.2	6.2	6.2	7.0	8.6	12.1	14.2	16.2	13.8	11.2	8.8	7.1		
		MCWB	4.3	4.5	4.5	4.8	6.3	9.2	11.8	14.2	11.4	8.9	7.7	5.5		
	5%	DB	5.1	5.0	5.1	6.2	7.6	11.0	12.9	14.8	12.8	10.1	7.7	5.2		
		MCWB	3.5	3.5	3.4	4.0	5.3	8.6	10.8	12.8	10.7	8.2	6.4	3.6		
	10%	DB	4.0	4.0	4.1	5.2	7.3	9.7	12.3	13.6	12.3	9.0	7.1	4.1		
		MCWB	2.6	2.5	2.4	3.4	5.0	7.8	10.3	11.6	10.2	7.3	5.6	2.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.4	6.4	6.8	8.1	10.9	14.8	16.9	14.2	10.6	9.5	7.5		
		MCDB	7.1	7.5	7.2	7.4	8.7	12.8	16.4	18.0	15.1	11.9	10.4	8.0		
	2%	WB	4.8	4.9	5.1	5.2	6.8	9.5	12.6	14.8	12.3	9.2	7.9	5.5		
		MCDB	6.1	5.9	6.2	6.5	7.9	11.2	14.1	16.1	13.2	10.6	8.5	6.8		
	5%	WB	3.9	3.9	3.9	4.5	5.9	8.5	11.3	13.3	11.3	8.2	6.9	4.1		
		MCDB	5.0	5.0	5.1	5.9	7.3	10.1	12.8	14.6	12.4	9.6	7.6	5.3		
	10%	WB	3.1	3.0	3.1	3.8	5.3	7.9	10.3	12.0	10.3	7.7	5.6	3.3		
		MCDB	4.0	4.0	4.2	5.2	6.8	9.1	11.9	13.0	11.7	8.8	7.0	4.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	3.8	3.9	4.1	3.9	3.8	4.2	4.4	4.4	4.4	4.0	4.2		
		MCDBR	3.9	4.2	4.7	4.9	4.7	5.8	5.9	6.0	5.2	5.1	5.1	4.6		
	5% WB	MCWBR	3.6	4.3	4.2	4.0	3.6	3.9	4.3	4.6	3.8	4.2	5.1	4.5		
		MCWBR	3.8	4.4	4.6	4.5	4.2	4.9	5.9	5.7	4.6	4.7	5.1	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.334	0.358	0.435	0.428	0.416	0.431	0.402	0.391	0.354	0.344	0.304		
		taud	2.318	2.311	2.293	2.041	2.133	2.260	2.264	2.366	2.291	2.310	2.195	2.289		
	Ebn at Noon		684	776	830	795	825	840	818	825	783	737	612	621		
		Edh at Noon	64	85	104	151	145	129	127	109	104	84	71	56		
(44) All-Sky Solar Radiation	RadAvg		0.71	1.30	2.48	3.67	3.90	3.91	3.75	3.26	2.71	1.86	0.88	0.56		
	RadStd		0.06	0.15	0.20	0.32	0.31	0.32	0.31	0.30	0.24	0.17	0.08	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page